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Exchange rate exposure and industry characteristics: evidence from Canada, Japan, and the USA

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This paper examines industry-level exchange rate exposures for Canada, Japan, and the USA. Measuring exposure by adding the change in the exchange rate to the domestic market model of industry portfolio returns, some industries in all three countries display significant exposures. Moreover, for each country, the exchange rate is important for explaining industry returns at the economy-wide level. To explore whether exchange rate exposures are systematically linked to the activities of the industries, we model exposure as a function of industry characteristics. For all three countries, the relation between exposure and industry characteristics is broadly consistent with economic theory. (JEL F3)

Exchange rate fluctuations can have a substantial impact on the profitability of domestic industries. Price changes caused by movements in the exchange rate may: (1) change the terms of competition with foreign firms for domestic exporters and import competitors; (2) alter input prices for industries that use internationally-priced inputs or firms that import for resale; and (3) change the value of assets denominated in foreign currencies. Because of this diverse set of

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influences, exchange rate movements should affect some industries differently than others.

While the effect of exchange rate fluctuations on an industry should depend critically on the industry's relation with the world economy, there are few inter-industry studies that document these different effects. Examples of such studies include Ceglowski (1989), which finds that a depreciation of the dollar significantly affects imports and exports in some US industries, and Goldberg (1990), which finds significant correlations between exchange rate movements and domestic investment for many US industries. Despite finding significant relations between the exchange rate and trade flows and investment, these studies, which focus only on the USA, have not related the effects of exchange rate changes to industry profitability and value.

Using industry portfolio returns for Canada, Japan, and the United States, this paper examines the relation between changes in exchange rates and industry values and explores whether industry characteristics, such as trade ratios, use of internationally-priced inputs, and foreign investments, systematically determine this relation. In the spirit of Adler and Dumas (1984), we identify an asset's exposure to exchange rate fluctuations as the correlation between the value of the asset and the exchange rate. Since the hypothesis of efficient markets suggests that changes in the market value of the firms in an industry reflect changes in current or expected conditions relevant for the profitability of that industry, we add the change in the exchange rate to the market model of returns as a measure of an industry's exchange rate exposure.

While the coefficient on the exchange rate variable in the agumented market model measures the industry's exchange rate exposure, this measure does not provide information on the determinants of the exposure. The exposure coefficient should depend on the interrelation of the exposures of the industry's activities. In particular, exposures may be large for an industry heavily involved in a single activity, such as exporting or importing, but they may be small for industries that undertake combinations of activities. For example, an appreciation of the home currency may have little effect on the value of an industry that both invests abroad and uses internationally-priced inputs even though the appreciation lowers the value of foreign investments and lowers the cost of such inputs. Since the interrelation of these activities should determine industry exchange rate exposure, we model exposure as a function of industry characteristics.

Prior research on exchange rate exposure, using similar methodology, has examined firm-level data. Jorion (1990) finds that a set of US multinationals has small (relative to their standard error) exchange rate exposures; however, he finds a positive relation between firms' exposure to exchange rate depreciation and the ratio of their foreign sales to total sales. For Canadian natural resource firms, Booth and Rotenberg (1990) find the surprising result that many of these firms benefit from appreciations of the Canadian dollar. They also find that firms' foreign sales, assets, and debt help explain their exposures. With industry-level rather than firm-level data, we examine a broader cross-section of the economy as well as different characteristics.

Section I of the paper sketches the theoretical relation between exchange rate exposure and industry characteristics. Section II discusses our methodology and presents our results. Our results suggest that all three countries show some evidence of significant industry exchange rate exposures. Furthermore, the data

support the idea that exchange rate exposure depends on industry characteristics: for all three countries, the relation between exchange rate changes and industry values varies systematically with industry characteristics in a manner broadly consistent with economic theory. Section III offers concluding remarks.

I. Exchange rate exposure and industry profits: theoretical relations

The standard definition of exchange rate exposure is a measure of the correlation between real asset values and real exchange rates. This definition does not imply a causal relation between changes in asset values and changes in exchange rates; instead, it allows asset values and exchange rates to be determined simultaneously by underlying factors in the economy. Despite this possible simultaneity, the partial equilibrium assumption that the exchange rate is exogenous to industry value is justifiable for a single industry: if the industry is a small part of a country's activity in the international economy, the exchange rate depends much more on events in other industries than on events in the industry under consideration. Even if the simultaneity were important for some industry, inter-industry differences in exposures should persist since the exposures reflect the contemporaneous impact of underlying factors on both exchange rate changes and industry value.

The correlation between an industry's profitability and changes in the value of the home currency should depend on what the industry does. The extent to which an industry exports or imports, the type of markets on which it obtains inputs, and its foreign investments all affect an industry's linkage with the international environment and, therefore, its exposure to exchange rate changes. Below, we consider how inter-industry differences in these activities should affect an industry's exposure to real exchange rate changes.

To begin with, changes in the exchange rate may affect non-traded goods industries differently than traded goods industries. Non-traded goods are goods for which high transportation costs prevent international trade. Macroeconomic models, such as Dornbusch (1974) and Gavin (1988), predict that the relative price changes of an appreciation of the home currency induce a shift of resources from traded to non-traded industries as long as capital is more sector specific than other inputs to production. This reallocation of resources causes the market value of capital in non-traded goods industries in the short run to rise relative to the market value of capital in traded goods industries. This suggests a positive relation between the value of non-traded goods industries and appreciations.

For traded goods, since the exchange rate is the relative price of domestic to foreign goods, its movements change relative input and output prices that affect an industry's current and future operating cash flows and, thereby, its value. Consider a country with an imperfectly competitive export sector, an import sector (where firms distribute imports), and an import-competing sector.[1] Initially, let all inputs to production be available from domestic markets that are insulated from international conditions. Now consider the differential response to a home-currency appreciation across the three sectors. An appreciation lowers the amount of home currency needed to purchase a unit of foreign currency, which *ceteris parabis* results in a lower home-currency price of foreign goods and a higher foreign-currency price of home goods.[2] In general, this helps the import sector and hurts the export and import-competing sectors. More specifically, the

appreciation reduces cash flows (measured in the home currency) of exporters because it causes some combination of a decrease in foreign demand and a lower price-cost margin, depending on the degree of exchange rate passthrough.[3] For importers, since the appreciation lowers their home-currency cost of goods, their cash flows increase through some combination of increased demand and higher price cost margins (depending on the degree of passthrough). At the same time, the increased price competitiveness of foreign imports results in some loss of demand and squeezed margins for the import-competing sector. A depreciation has the opposite effects on industry profitability.

Changing the assumption of isolated domestic input markets to allow for internationally-priced inputs highlights another possible effect of the exchange rate on cash flows of both traded and non-traded industries. The term internationally-priced input markets includes both inputs which are imported and inputs obtained domestically whose price is determined on world markets. Assuming input markets are competitive, an appreciation of the home currency lowers the home-currency price of internationally-priced inputs, so production costs fall and industry profitability rises. Similarly, a depreciation increases the home currency price of these inputs, increasing costs and decreasing profitability.

Finally, exchange rate changes directly affect the value of foreign denominated assets through the translation of values from one currency to another. For example, firms with foreign investments have current and future cash flows denominated in foreign currency and the home-currency value of this stream of cash flows depends on the exchange rate. In most cases, a depreciation of the home currency increases the value of industries with net foreign denominated assets, while an appreciation decreases the value of these industries.[4]

Table 1 summarizes these effects of exchange rate changes on different activities' profitability. The impact of exchange rate movements on an industry's profitability is more difficult to determine when the industry undertakes activities with different exposures.

TABLE 1. The effects of an appreciation of the home currency on the value of industries involved in different activities.

Activity	Sign of effect
Non-traded good producer	(+)
Exporter	(−)
Importer	(+)
Import competitor	(−)
User of internationally-priced inputs	(+)
Foreign investor	(−)

Notes: The entries in the table represent how an appreciation of the home currency affects the domestic industry value. A plus sign indicates that an industry involved in that activity gains from an appreciation; a negative sign indicates the opposite. To the extent that an industry is involved in more than one activity, the industry's exposure is less apparent.

II. Tests and results

While the theories above relate industry value, industry characteristics, and changes in real exchange rates, our empirical analysis uses the nominal exchange rate. There are two reasons for this substitution. First, using the real exchange rate would assume that financial markets instantaneously observe the inflation rates that are necessary for calculating the real exchange rate. Since nominal exchange rates are readily observable, it is less demanding to assume that the markets correctly measure nominal exchange rates. Second, over the periods we consider, changes in nominal and real exchange rates are highly correlated. Using the consumer price indices, the correlations between monthly changes in the nominal and real G-7 trade-weighted exchange rates over the sample period are 0.97 for the USA, 0.95 for Canada, and 0.98 for Japan.

To measure the exposure of industry portfolios to exchange rate fluctuations, we add the contemporaneous change in the exchange rate to the domestic market model used to explain stock price movements. Under the hypothesis of efficient equity markets, movements in the stock market value of an industry should reflect the effect of any news, such as unexpected changes in exchange rates, on future profits. Since previous work in international finance, most notably Meese and Rogoff (1983), shows that changes in the nominal exchange rate are virtually unforecastable, we use the percentage change in the nominal exchange rate as a proxy for the innovations or 'news' in the series.[5] This property suggests a large permanent component in exchange rate movements, so current exchange rate movements also affect future levels of the exchange rate. For each country, the exchange rate variable is a trade-weighted nominal exchange rate (using weights from the International Monetary Fund's Multilateral Exchange Rate Model (MERM)) among the other six G-7 countries.[6] This measure provides an effective exchange rate for each currency against the other six currencies. The exchange rate is a month-end to month-end measure making its return comparable to the portfolio returns. We define the exchange rate such that an increase in the variable represents an appreciation of the home currency.

Our tests measure the effect of exchange rate news on *ex post* stock price movements. Since we are concerned only with whether exchange rate movements affect *ex post* returns as opposed to expected (*ex ante*) returns, our tests differ from tests of whether the exchange rate is a 'priced' factor in stock market returns. Previous research, such as Dominguez (1992, ch. 3) and Hamao (1988), includes the exchange rate as an observable factor in asset pricing models with mixed results. Regardless of whether exchange rate exposure is a priced factor that influences expected returns, exchange rate changes can systematically affect realized returns since they provide information about economic conditions.

II.A. Industry regressions

To measure individual industry exchange rate exposures, we estimate the following equation for each industry in the three countries:

$$\langle 1 \rangle \qquad (R_{i,t} - rf_t) = B_{0,i} + B_{1,i} \cdot (R_{m,t} - rf_t) + B_{2,i} \cdot PCXR_t + \varepsilon_{i,t},$$

where $R_{i,t}$ is the return on industry portfolio i in month t, rf_t is the risk-free rate of return in month t, $R_{m,t}$ is the return to the national stock market in month t,

$PCXR_t$ is the percentage change in the trade-weighted nominal exchange rate in month t, $B_{0,i}$, $B_{1,i}$, and $B_{2,i}$ are parameters and $\varepsilon_{i,t}$ is the residual for the return to industry i in month t. As in the market model, where the market beta (the analogue to $B_{1,i}$) measures the industry's exposure to changes in the overall stock market index, the exchange rate coefficient, $B_{2,i}$, measures the industry's exposure to exchange rate fluctuations. More precisely, $B_{2,i}$ measures the industry's exposure to exchange rate appreciations independent of the overall market's exposure to appreciations. Positive $B_{2,i}$s indicate that the industry benefits from an appreciation of the home currency.

For Canada and the USA, we estimate equation ⟨1⟩ using seemingly unrelated regressions (SUR) over the ten-year period January 1979 to December 1988. In this case, SUR is the equivalent of generalized least squares (GLS), which accounts for contemporaneous correlation of the error terms across the industries in both Canada and the USA.[7] For Japan, we estimate equation ⟨1⟩ by ordinary least squares (OLS) on each industry over the period September 1983 to December 1988;[8] system estimation techniques do not offer any econometric advantage since each industry equation has the same regressors. We are unable to run a multi-country system regression over the shorter Japanese sample period because feasibility of system techniques requires that the number of time periods be much larger than the number of cross-sections.[9]

Panels A, B, and C of Table 2 present industry exchange rate exposures from equation ⟨1⟩ for the USA, Canada, and Japan. In all three countries, well less than half of the industries have exposures that are statistically significant at the 10 per cent level: for the USA, 11 of 39 industries have significant exchange rate exposures (28 per cent); for Canada, four of 19 industries have significant exposures (21 per cent); and for Japan, seven of 20 industries have significant exposures (35 per cent). For the USA, the Metal Mining, Heavy Construction, Petroleum Refining, Wholesale and Durable Trade, and Business Services industries have significant negative exposures and the Apparel, Transport Equipment, Air Transport, Motor Freight Transportation, General Merchandise Stores, and Miscellaneous Retail have significant positive exposures. For Canada, the Metal Mining and the Paper and Forest Products industries have significant negative exposures, and the Department Stores and Transportation industries have significant positive exposures. Finally, for Japan, the Chemicals, Electric Machinery, and Precision Instruments industries have significant negative exposures, and the Construction, Oil and Coal, Land Transport, and Services industries have significant positive exposures.

While many industries do not have significant exposures, it is not true that the exchange rate is an insignificant variable for explaining industry returns at the overall economy level. The test of whether exchange rate changes help explain economy-wide industry returns is the joint test that all industry exchange rate exposures are zero for each country. The results of this test, displayed in Table 2, are statistically significant for all three countries suggesting that, at an economy-wide level, exchange rate fluctuations help explain industry returns. For the USA and Canada, we reject that the set of industry exchange rate exposures are zero at the 5 per cent level, and for Japan we reject this hypothesis at the 1 per cent level.

One hypothesis, examined below, of why so many industries have insignficant exposures is that industries undertake a variety of different activities with different

TABLE 2. Industry-level exchange rate exposures.

Panel A: United States

$$(R_{i,t} - rf_t) = B_{0,i} + B_{1,t} \cdot (R_{m,t} - rf_t) + B_{2,i} \cdot PCXR_i + \varepsilon_{i,t}$$

No.	SIC	Industry name	$B_{2,i}$	Standard error
1	10	Metal Mining	−0.5795	(0.2152)**
2	13	Oil and Gas Extraction	−0.1567	(0.1775)
3	15	Building Construction	−0.0805	(0.1760)
4	16	Heavy Construction Other than Buildings	−0.3320	(0.1731)*
5	20	Food and Kindred Products	−0.1073	(0.0705)
6	21	Tobacco	−0.0448	(0.1359)
7	22	Textile Mill Products	0.0767	(0.1228)
8	23	Apparel & Other Clothes	0.3043	(0.1146)***
9	24	Lumber and Wood	−0.0653	(0.1610)
10	25	Furniture and Fixtures	0.1376	(0.1400)
11	26	Paper and Allied Products	−0.1332	(0.0937)
12	27	Printing and Publishing	−0.0834	(0.0800)
13	28	Chemicals	−0.0603	(0.0525)
14	29	Petroleum Refining	−0.2958	(0.1384)**
15	30	Rubber and Misc. Plastics	−0.0320	(0.0966)
16	31	Leather and Leather Products	0.2184	(0.1453)
17	32	Stone, Clay and Concrete	0.0084	(0.0720)
18	33	Primary Metals	0.0253	(0.0898)
19	34	Fabricated Metal Products, except Machines	−0.0301	(0.0534)
20	35	Industrial Machinery & Computer Equipment	−0.0159	(0.0733)
21	36	Electronic Equipment, except Computers	−0.0451	(0.0657)
22	37	Transport Equipment	0.2039	(0.0669)***
23	38	Instruments	−0.0138	(0.0819)
24	40	Railroad Transport	−0.0218	(0.1158)
25	42	Motor Freight Transportation	0.4547	(0.1888)**
26	44	Water Transport	−0.2139	(0.1795)
27	45	Air Transport	0.3464	(0.1781)*
28	48	Communication	−0.0541	(0.0675)
29	49	Public Utilities	−0.1129	(0.0865)
30	50	Wholesale Trade, Durable Goods	−0.1362	(0.0652)**
31	53	General Merchandise Stores	0.3045	(0.1128)***
32	54	Food Stores	0.1171	(0.1187)
33	56	Apparel & Accessory Stores	0.1187	(0.1608)
34	58	Eating and Drinking Places	0.0529	(0.1607)
35	59	Miscellaneous Retail	0.2401	(0.0896)***
36	65	Real Estate	0.0708	(0.1113)
37	70	Hotels	0.1733	(0.1331)
38	73	Business Services	−0.2345	(0.0909)**
39	78	Motion Pictures	0.0273	(0.1418)

F-test = 1.4652 Significance = 0.0309
(H_0: All $B_{2,i} = 0$)

TABLE 2. Industry-level exchange rate exposures (cont.)

Panel B: Canada

$$(R_{i,t} - rf_t) = B_{0,i} + B_{1,i} \cdot (R_{m,t} - rf_t) + B_{2,i} \cdot PCXR_t + \varepsilon_{i,t}$$

No.	Industry name	$B_{2,i}$	Standard error
1	Cement and Concrete	−0.0047	(0.3202)
2	Chemicals	−0.3242	(0.3032)
3	Communications and Media	−0.2339	(0.2128)
4	Department Stores	0.9152	(0.3337)***
5	Electric Devices & Electronic Companies	−0.0033	(0.4349)
6	Food Processing	0.2455	(0.2093)
7	Food Stores	0.0309	(0.2532)
8	Machinery	0.5077	(0.5294)
9	Metal Mining	−0.9731	(0.2868)***
10	Metal Fabrication	0.3111	(0.2818)
11	Oil and Gas	0.3685	(0.2723)
12	Paper and Forest Products	−0.6169	(0.2565)**
13	Publishing and Printing	−0.2876	(0.2416)
14	Steel	0.1135	(0.2118)
15	Tobacco	−0.0132	(0.3197)
16	Transportation Equipment	0.3225	(0.3468)
17	Transportation	1.2965	(0.4683)***
18	Uranium and Coal	0.1193	(0.3265)
19	Utilities	0.1132	(0.1724)

F-test = 1.8088 Significance = 0.0169
(H_0: All $B_{2,i} = 0$)

exposures. Another reason why some exposures are not statistically significant is that firms may reduce their exposures through financial hedging using forwards, futures, options, and swaps. While such hedging does not affect cash flows from real operations, it creates a cash flow that reduces the correlation between total cash flows and the exchange rate. The amount of hedging within an industry may be endogenous since industries with inherently large exposures are the most likely to hedge through financial markets. Complete cash-flow hedging by an industry is theoretically possible (though not necessarily optimal, see Jacque, 1978), but our tests assume that hedging is incomplete. While financial hedging may affect our measures of exposure, we cannot test for the importance of financial hedging because data on hedging by industries are not available.

Before attempting to relate exposures to industry characteristics, the multinational nature of our data allows some comparisons of the industry-level exposures across the three countries. If, as suggested by open-economy macroeconomics, small and open economies are more sensitive to changes in international conditions, then one would expect the dispersion of industry-level exchange rate exposures to vary systematically with the size and openness of the economy: the smaller and more open the economy, the larger the inter-industry differences in exchange rate exposures. The results in Table 2 support these

TABLE 2. Industry-level exchange rate exposures (cont.)

Panel C: Japan

$$(R_{i,t} = rf_t) = B_{0,i} + B_{1,i} \cdot (R_{m,t} - rf_t) + B_{2,i} \cdot PCXR_t + \varepsilon_{i,t}$$

No.	Industry name	$B_{2,i}$	Standard error
1	Mining	−0.3316	(0.2803)
2	Construction	0.6996	(0.3035)**
3	Food and Food Products	0.0237	(0.2294)
4	Textiles	−0.2878	(0.2279)
5	Pulp and Paper	0.1927	(0.3019)
6	Chemicals	−0.2998	(0.1230)**
7	Oil and Coal Products	0.8482	(0.3348)**
8	Rubber Products	−0.2963	(0.3261)
9	Glass and Ceramics	−0.0666	(0.1480)
10	Iron and Steel	−0.0178	(0.2522)
11	Non-ferrous Metals	0.2170	(0.1955)
12	Machinery	−0.0866	(0.1767)
13	Electric Machinery	−0.7951	(0.3171)**
14	Transport Equipment	−0.0199	(0.2812)
15	Precision Instruments	−0.7366	(0.2498)***
16	Commerce	0.0224	(0.1679)
17	Land Transport	0.5325	(0.3045)*
18	Communications	0.0934	(0.3244)
19	Electric Power	0.8445	(0.5242)
20	Services	0.5649	(0.2967)*

F-test = 2.6267 Significance = 0.001
(H_0: All $B_{2,i} = 0$)

Notes: For the USA and Canada, data are monthly from 79:1 to 88:12, and the US and Canadian regressions are estimated as a system using SUR. For Japan, the data are monthly from 83:9 to 88:12 and the regression is estimated using OLS. For Panel A, the second column is the two-digit SIC code for the industry. For each country, $R_{i,t}$ is the return to industry i, $R_{m,t}$ is the return to the domestic market, $PCXR_t$ is the percentage change in a trade-weighted value of the domestic currency. The estimate of the coefficient $B_{2,i}$ is the exchange rate exposure of industry i. The industry constants, $B_{0,i}$, and market betas, $B_{1,i}$, are not reported for space considerations. (*) represents significance at the 10 per cent level, (**) at the 5 per cent level, and (***) at the 1 per cent level.

open-economy hypotheses.[10] The variance of the $B_{2,i}$, coefficients across industries is larger for Canada and Japan than for the USA. The USA, the largest and least open of three economies with a trade flow to GNP ratio of 0.17, has a variance of estimated exchange rate exposures of 0.04.[11] Japan, with a GNP approximately half that of the USA and a trade flow to GNP ratio of 0.22, has an exchange rate exposure variance of 0.21. Canada, the smallest and most open economy with GNP less than one-tenth of that of the USA and a total trade flow to GNP ratio of 0.55, has a variance of exchange rate exposures of 0.26.[12] While the dispersion of industry exchange rate exposures suggests that industries in smaller and more open economies are likely to be more highly exposed to exchange rate fluctuations, this result should be interpreted with caution as many of the individual exposures are not statistically different from zero.

II.B. Exposure and industry characteristics

The joint tests above suggest that exchange rate changes help explain industry returns in all three countries, but many individual industries do not have significant exchange rate exposures. Modeling exchange rate exposure as a function of industry characteristics, such as importing and foreign investment, has two advantages over examining industry-level exposures. First, economic theory suggests that cross-sectional differences in industry-level exposure should depend systematically on industry characteristics. Second, the fact that many industries pursue activities that have different impacts on their overall exposure might provide one reason why more than half of the industries in Table 2 have statistically insignificant exchange rate exposures. As outlined in Table 1, different activities imply particular exposures to exchange rate fluctuations; however, involvement in multiple activities makes it more difficult to predict an industry's exchange rate exposure.

Testing the hypothesis that industry characteristics determine exchange rate exposures requires measures of each industry's involvement in the activities listed in Table 1. Data regarding the activities of the industries in each country are obtained from various sources.[13] To differentiate between traded and non-traded goods, we create a dummy variable for non-traded industries (NONTRA, 0 = traded goods, 1 = non-traded goods). An export ratio (PCEXP), defined as exports over total domestic production, captures the degree to which industries export. Similarly, an import ratio (PCIMP), defined as imports over total domestic consumption, measures the import penetration of an industry. Unfortunately, this variable does not distinguish between importing for resale by the domestic industry, which implies gains to the industries from an appreciation, and imports by foreign exporters, which implies losses to competing domestic industries from an appreciation. As a proxy for the importance of internationally-priced inputs, we use the percentage of final value spent on oil and coal products taken from input tables for each country (RAWMAT); however, since these products are priced in US dollars on the international market, it is not clear that this variable should affect US industry exposures. For the USA and Japan, we use the ratio of an industry's foreign assets to its total assets (PCFDI) as a measure of the importance of its foreign investment holdings. Unfortunately, similar data for Canada are not available.

To determine the empirical relation of these characteristics to an industry's exposure to exchange rate movements, we model exposure as a linear function of industry characteristics:

$$\langle 2 \rangle \qquad B_{2,i} = \gamma_0 + \sum_{k=1}^{n} \gamma_k \cdot C_{k,i}, \qquad i = 1, \ldots, I,$$

where $\{C_{k,i}, k = 1, \ldots, n\}$ is the set of characteristics for industry i.[14] Substituting equation ⟨2⟩ into equation ⟨1⟩, which allows for more efficient estimation, yields:

$$\langle 3 \rangle \qquad (R_{i,t} - rf_t) = B_{0,i} + B_{1,i} \cdot (R_{m,t} - rf_t) + \gamma_0 \cdot PCXR_t + \sum_{k=1}^{n} \gamma_k \cdot (C_{k,t} \cdot PCXR_t) + \varepsilon_{i,t}, \quad i = 1, \ldots, I.$$

We estimate this interacted regression in a similar fashion to equation ⟨1⟩. Generalized least squares estimation corrects for contemporaneous correlations among the industries' residuals. The US and Canadian industries are estimated together as a system, while the Japanese industries are estimated separately because of data limitations. The interacted specification allows tests of the joint hypothesis, which we refer to as the industry characteristic exposure (ICE) hypothesis, that (i) exchange rate exposure affects industry returns, and (ii) exchange rate exposure depends on industry characteristics in a linear relation. Our null hypothesis is that exchange rate exposure is not important to industry returns, which is the joint test that $\gamma_k = 0$ for $k = 0, 1, \ldots, n$.

Table 3 contains estimates of the industry characteristic coefficients from equation ⟨3⟩. The new variables, created by interacting the cross-sectional industry characteristics with the exchange rate series ($C_{k,i} \cdot PCXR$), are represented by the industry characteristic variable name appended with an 'X' to emphasize the interaction with the exchange rate.

For the USA (Panel A), estimates of most of the coefficients on the industry characteristics have signs that are consistent with the theoretical implications of Table 1; however, several of the coefficients are not estimated precisely. The estimate of the coefficient on the non-traded dummy (NONTRAX) is close to zero (−0.026) and indicates that US non-tradable goods industries show no statistical relation to movements of the dollar.[15] The estimated coefficient on the export ratio (PCEXPX) is not statistically significant, but, as expected, its point estimate is negative (−0.235), suggesting that industries that export (more heavily) benefit from a depreciation of the dollar. The estimate of the coefficient on the import ratio (PCIMPX) is positive (0.221) but is not statistically different from zero. The positive coefficient implies that an appreciation increases the value of importing industries, suggesting that the import ratio predominantly reflects imports for resale. The estimated coefficient on internationally-priced inputs (RAWMATX) is negative, −0.719, and is significant at the 1 per cent level.[16] Contrary to the expectation that this variable should have no effect for the US industries, this result indicates that US industries that rely heavily on oil and coal gain from a depreciation of the dollar. The estimate of the coefficient on foreign direct investment (PCFDIX) is negative, −0.715, and significant at the 1 per cent level. Consistent with the analysis above, this implies industries with foreign denominated assets suffer from an appreciation of the dollar. Finally, for the USA we reject our null hypothesis at the 1 per cent level; thus, the results support the industry characteristic exposure (ICE) hypothesis that exchange rate exposure depends on industry characteristics.

For Canada (Panel B), the signs of the parameter estimates match the theoretical implications from Table 1. The estimate of the coefficient for NONTRAX is positive, 0.306, and significant at the 1 per cent level. This indicates that Canadian non-traded industries, unlike US non-traded industries, gain relative to traded good industries with an appreciation of the domestic currency. The estimate of the coefficient for PCEXPX is negative, −0.933, and is statistically significant at the 5 per cent level. The estimated coefficient for PCIMPX is positive, 1.332, and significant at the 1 per cent level, implying that industries with large import penetration ratios import for domestic sale rather than face import competition. The estimate of the coefficient for RAWMATX is large and positive (1.414) but is not significantly different from zero, suggesting that while,

TABLE 3. Interacted industry characteristic regressions.

$$(R_{i,t} - rf_t) = B_{0,i} + B_{1,i}\cdot(R_{m,t} - rf_t) + \gamma_0\cdot PCXR_t + \sum_{k=1}^{n} \gamma_k\cdot(C_{k,i}\cdot PCXR_t) + \varepsilon_{i,t}$$

$$i = 1, \ldots, I$$

CONSTANTX γ_0	NONTRAX γ_1	PCEXPX γ_2	PCIMPX γ_3	RAWMATX γ_4	PCFDIX γ_5
A. USA					
0.1040	−0.0256	−0.2350	0.2207	−0.7193	−0.7154
(0.0423)	(0.0361)	(0.3085)	(0.2159)	(0.2401)	(0.2844)
0.016	0.478	0.445	0.320	0.002	0.010
	F-statistic = 4.1376		Significance level = 0.004		
	(H_0: $\gamma_0 = \gamma_1 = \gamma_2 = \gamma_3 = \gamma_4 = \gamma_5 = 0$)				
B. Canada					
−0.1193	0.3060	−0.9329	1.3322	1.4137	—
(0.1275)	(0.7000)	(0.4686)	(0.4805)	(1.7211)	
0.344	0.000	0.045	0.007	0.403	—
	F-statistic = 5.1398		Significance level = 0.001		
	(H_0: $\gamma_0 = \gamma_1 = \gamma_2 = \gamma_3 = \gamma_4 = 0$)				
C. Japan					
−0.0961	0.3945	−0.7746	0.9215	0.2964	−2.1777
(0.0680)	(0.0980)	(0.3504)	(0.9943)	(0.4746)	(1.1976)
0.158	0.001	0.027	0.354	0.532	0.0693
	F-statistic = 6.5196		Significance level = 0.001		
	(H_0: $\gamma_0 = \gamma_1 = \gamma_2 = \gamma_3 = \gamma_4 = \gamma_5 = 0$)				

Notes: Below the coefficients, the standard errors are in parentheses followed by the significance value. Estimates of $B_{0,i}$ and $B_{1,i}$ are not reported. The variables match those in Table 2, with the addition of the industry characteristics. NONTRAX is the interacted non-traded goods dummy; PCEXPX is interacted per cent exports; PCIMPX is interacted import penetration; RAWMATX is interacted internationally-priced inputs; and PCFDIX is interacted foreign investment. The sample period for the USA and Canada is 79:1 to 88:12 and for Japan is 83:9 to 88:12. The USA and Canada are estimated as a system; Japan is estimated by itself due to the smaller sample.

on average, users of oil and coal in Canada gain from an appreciation of the Canadian dollar, this result is not uniform across industries. For Canada, as for the USA, the data support the ICE hypothesis that exchange rate exposures are a linear function of industry characteristics by rejecting the null hypothesis at the 1 per cent level.

For Japan, as with Canada, the coefficients on the industry characteristics are consistent with the theoretical predictions of Table 1. The estimate of the coefficient on NONTRAX is positive, 0.395 and is significant at the 1 per cent level, suggesting that exchange rate appreciations help Japanese non-traded industries. The estimate of the coefficient on PCEXPX is both negative, −0.775,

and significant at the 5 per cent level. The estimate of the coefficient on PCIMPX, while positive and large (0.922), is not statistically different from zero; however, the positive value suggests that the import variable is measuring imports for resale by industries as opposed to import competition. The estimate of the RAWMATX coefficient is positive (0.296) but is not statistically different from zero. As with the USA, the estimate of the coefficient on PCFDIX is negative, −2.178, and significant at the 10 per cent level, consistent with a depreciation increasing the value of foreign operations. Finally, as for the other two countries, rejecting the null hypothesis at the 1 per cent level, the results support the ICE hypothesis that industry characteristics are systemically related to exchange rate exposures.

While the statistical significance of the interacted regressions suggests that exposures in each country depend on industry characteristics, these tests do not reveal how well the linear specification explains the estimated industry-level exposures from equation ⟨1⟩ (displayed in Table 2). To determine how well industry characteristics explain industry-level exposures, we test whether the exposures predicted by equation ⟨2⟩, based upon industry characteristics and estimated parameters, are significantly correlated with the exposures from Table 2. If these two measures are significantly correlated, then the linear specification of industry characteristics is useful for explaining industry exposures. Moreover, a significant correlation may suggest that some of the small (or statistically insignificant) coefficients in Table 2 arise because industries undertake activities with offsetting exposures.

For all three countries, the industry exchange rate exposures predicted by the linear combination of industry characteristics are significantly correlated with the estimated industry exchange rate exposures from the augmented market model.[17] The two sets of exposure coefficients are most highly correlated for Japan: the correlation between the predicted (as a function of industry characteristics) and estimated (from simple augmented market model) exposures is 0.80 and is significant at the 1 per cent level. For Canada, even without the advantage of a foreign investment variable, the correlation between the two sets of exposures is 0.49 and is significant at the 5 per cent level. For the USA, the correlation between the exposures predicted from the linear specification and the exposures estimated in Table 2 is 0.34 and is significant at the 5 per cent level. These results indicate that the industry characteristics possess a reasonable degree of explanatory power for the industry exposures in each of the three countries and the insignificant industry-level exposures for more than half of the industries in each country might result from the interaction of their exposures to different activities.

III. Conclusion

This paper presents evidence on the effect of exchange rate fluctuations on industry value using data on stock market returns to industry portfolios. We study a broad spectrum of industries encompassing traded and non-traded, manufacturing and service industries in Canada, Japan, and the USA. Besides allowing estimation of industry-level exchange rate exposures, our broad set of industries allows us to examine determinants of exchange rate exposures.

To identify the effect of exchange rate movements on industry profitability, we use an augmented market model. This specification estimates exchange rate exposures at the industry level by adding the innovation in a trade-weighted exchange rate to the market model for industry portfolios in each country. For all three countries, we find similar results: between 20 and 35 per cent of industries have statistically significant exchange rate exposures and exchange rate fluctuations help determine industry returns at an economy-wide level. The data also suggest that the impact of exchange rate movements on industry returns is larger for Canada and Japan than for the USA, which is consistent with the exchange rate having a larger impact on smaller and more internationally-oriented economies.

By specifying exchange rate exposures as a function of industry characteristics, we are able to determine whether there is a systematic relation among industry exposures and industry activities. We decompose industry exchange rate exposure into a linear relation of a non-traded industry dummy variable, an export ratio, an import penetration ratio, a measure of the reliance on internationally-priced inputs, and the ratio of foreign assets to total assets. From this specification, the results for all three countries suggest that these characteristics influence an industry's exchange rate exposure in a manner which is broadly consistent with economic theory. Except in the USA, the results indicate that non-traded goods industries gain from an appreciation of the home currency. The results also indicate that industry export ratios are associated with negative exchange rate exposures and industry import ratios are associated with positive exposures. The use of internationally-priced inputs results in positive exposures in Canada and Japan, but results in a paradoxical negative exposure in the USA. In both Japan and the USA, foreign denominated assets are significantly related with negative exposures to exchange rate changes. Further tests indicate that the predicted exposures from the simple linear specification are significantly correlated with estimated industry-level exposures for all three countries.

Data appendix

Stock market data: For the United States, stock return data are from the Center for Research in Security Price (CRSP) tapes for firms on the NYSE and AMEX. We create industry portfolios by two-digit Standard Industrial Classification (SIC) code groups. Industry returns are the equally-weighted average of dividend exclusive returns of firms in the industry. Only industries with no less than four firms active at any time during the sample are included. The market return is the dividend exclusive return for the equally-weighted market portfolio from CRSP.

For Canada, the stock market data are monthly stock price indices for firms traded on the Toronto Stock Exchange from the DBCAND Database of Data Resources Inc. Industry classifications are from the Toronto Stock Exchange. The price indices exclude dividends. Returns are calculated as the percentage change in the price index over the month. The market variable is the Toronto Stock Exchange 300 Index (same source).

For Japan, stock market data are value-weighted industry portfolio price indices for firms in the Nikkei 500 from the Nikkei Needs Database. The

end-of-month price indices exclude dividends. Returns are the percentage change over the month. The market return is the return on the Nikkei 500 Index.

Risk-free interest rates: For the USA and Canada the risk-free rate is the one-month return on a three-month treasury bill from the IMF's International Financial Statistics (IFS) tape. For Japan, since treasury bill rates are unavailable, the one-month return on a three-month money-market rate is from IFS tapes.

Exchange rates: For all three countries, we calculate an end-of-month effective exchange rate against the six other members of the G-7 (Canada, France, Germany, Italy, Japan, the UK, and the USA). Our method and relative weights match those used by the IMF for its Multilateral Exchange Rate Model (MERM) exchange rate. The exchange rate data are series code ag (end-of-month) from the IFS tapes. For relative weights, see the MERM table of weights available in the *IFS Supplement on Exchange Rates*, IMF, 1985, Appendix IV, p. 143. The exchange rate is an index in the form of units of foreign currency per unit of home currency. In this form an appreciation of the domestic currency is represented by an increase in the index.

Industry characteristic data: The non-traded dummy equals one if the industry produces a non-tradeable good or service. For the USA, the non-traded industries are industry numbers 15, 16, and 40–78. The per cent export data (dollar value of exports over production) and import penetration (dollar value of imports over consumption) for the manufacturing industries come from *The Statistical Abstract of the United States 1987*, Tables 1320 and 1321 for 1984. For the mining industries, similar data are obtained from *The Bureau of Census, Census of Mining 1977* for 1976. By definition these variables are zero for non-traded industries. The raw material variable is measured as percentage spent on petroleum and coal products for every dollar of output from input coefficient tables in 'Summary of Input–Output Tables for the US Economy: 1976, 1978 and 1979,' Bureau of Economic Analysis, Staff Papers, May, 1983, pp. 60–67. The data used are for 1978. The foreign direct investment variable, measured as the ratio of foreign denominated assets to total assets of the parent industry, is from *U.S. Direct Investment Abroad: 1982 Benchmark Survey Data*, US Dept. of Commerce, Bureau of Economic Analysis, 1985, pp. 47, 53.

For Canada, the non-tradeable industries are communications, department stores, food stores, transportation, and utilities. The import and export variables, for 1986, are from the *Canada Yearbook 1989*. The raw materials data are from *The Input–Output Structure of the Canadian Economy: 1970–1977*, Statistics Canada, 1981, pp. 225–246. Data are for 1977. We assign two industries with missing raw materials data the mean value for the other industries.

For Japan, the non-traded industries are construction, commerce, land transport, communications, electric power, and services. Data for per cent exports and import penetration come from *1988 Japan Statistical Yearbook*, Statistics Bureau, Management and Coordination Agency (ed.) Japan Statistical Association 1988, pp. 219, 340. When industry production was not available, sales figures were used. Consumption is defined as production plus imports. All data are for 1986. Raw material figures were not available for 1986, so we use 1980 figures. These come from *1980 Input–Output Tables, English Summary*, Administrative Management Agency, Government of Japan, March 1984, pp.

360–422. Foreign investment data are from the Japanese Ministry of Finance, *Financial Statistics of Japan 1989.*

Notes

1. For more formal models of these effects, see, for example, Shapiro (1975) or Levi (1990).
2. In this example, we assume that foreign firms do not act strategically; that is, they keep prices fixed in their own currency. Strategic price changes by foreign firms may dampen these effects.
3. Passthrough refers to the percentage change in the market price of a tradeable that occurs when the exchange rate changes. Zero passthrough implies that import prices do not change in the importer's currency and that the exporter absorbs the entire change in the exchange rate; complete passthrough implies that import prices change one-for-one with the exchange rate. We assume that passthrough for all sectors is strictly between zero and one.
4. The exposure of foreign assets will depend on whether the local currency value of the assets changes as the exchange rate changes. In certain cases if the foreign currency value of assets falls as a result of a home currency depreciation, the exposure to foreign assets may become positive.
5. Mussa (1979) argues that over 90 per cent of month-to-month US exchange rate changes are unpredictable. Furthermore, he also shows that changes in forward exchange rates and spot exchange rates are highly correlated so that changes in spot rates provide the same information as changes in forward rates on expected values of future exchange rates.
6. For a detailed description of all the data, see the data appendix.
7. The correlation between parameter estimates for the B_2s from SUR and OLS is 0.931 for the USA and 0.933 for Canada, suggesting that the system technique does not substantially influence the parameter estimates.
8. Unfortunately, available Japanese data from our database over the longer sample period consist of monthly average prices for industry portfolios. Month-end to month-end Japanese industry returns are available only for this shorter period.
9. Work on asset pricing models also suggests that the small sample properties of test statistics may differ substantially from their asymptotic distribution when the number of cross-sections is not substantially smaller than the number of time-series observations, see, *e.g.*, Shanken (1985).
10. These tests are not corrected for possible influences of differential estimation error of the exchange rate exposures.
11. The trade flow to GNP ratios are calculated as the sum of exports and imports to GNP. These data are from the *International Financial Statistics Yearbook* for 1986.
12. Pair-wise tests of the equality of these variances, using a Folded form F-test, confirm, at the 1 per cent significance level, that the US exposures have less variance than those for either Canada of Japan. The hypothesis that the Canadian and Japanese exposures have the same variance cannot be rejected.
13. Data on industry characteristics are not available on an annual basis in each of the three countries; therefore, point estimates of the characteristics are used. The observations are taken for a year as close to the middle of the sample period as possible. See the data appendix for details.
14. The linear specification without an error term follows Auerbach's (1983) work on dividend yields and firm characteristics. With this specification, the error structure of the interacted equation is operational.
15. Our classification of some industries as non-traded is somewhat arbitrary. For example, the motion picture industry (which we classify as non-traded) does have some international trade; however, measuring the amount is difficult (*e.g.*, films may be distributed internationally through a license or royalty arrangement). To check the sensitivity of our results to several cases of arbitrary classifications of industries, we estimate the US regression excluding those industries for which the non-traded classification seems arbitrary (Water Transport, Air Transport, Hotels, Business Services, and Motion Pictures). The results without these industries are similar to those reported in Table 3 (available from the authors).

16. The RAWMATX variable is included in the US regression for comparability with the other countries. Regressions omitting this variable for the USA yield similar results and significance levels for the coefficients and tests, with the exception that the constant is no longer significant.
17. This test does not account for the estimation error in the coefficients of the industry characteristics.

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